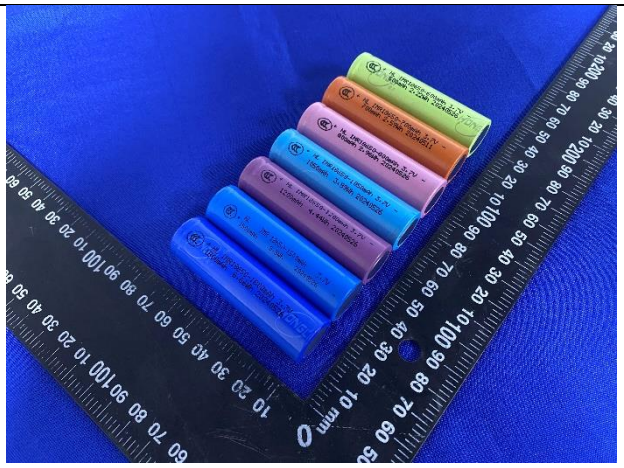
Prüfbericht-Nr.: *Test report no.*:	**CN24F6CW 001**	**Auftrags-Nr.:** *Order no.*:	168505406	Seite 1 von 34 *Page 1 of 34*
Kunden-Referenz-Nr.: *Client reference no.*:	N/A	**Auftragsdatum:** *Order date:*	2024-08-13	
Auftraggeber: *Client*:	Xinxiang Hongli Supply Source Technology Co.,Ltd. West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China			
Prüfgegenstand: *Test item*:	Cylindrical Lithium-ion Rechargeable Cell			
Bezeichnung / Typ-Nr.: *Identification / Type no.*:	See table 4 on page 6			
Auftrags-Inhalt: *Order content*:	TÜVus mark approval			
Prüfgrundlage: *Test specification*:	UL 1642:2020 R10.22			

Wareneingangsdatum: *Date of sample receipt*:	2024-08-13
Prüfmuster-Nr.: *Test sample no:*	SN240813012C001 ~ C115 SN240813013C001 ~ C115 SN240813014C001 ~ C115
Prüfzeitraum: *Testing period*:	2024-08-13 to 2024-09-19
Ort der Prüfung: *Place of testing*:	CMC Testing Intemational(Shenzhen) Co., Ltd.
Prüflaboratorium: *Testing laboratory:*	TÜV Rheinland (Shenzhen) Co., Ltd.
Prüfergebnis*: *Test result*:*	Pass



geprüft von: tested by: Revan Dai	X Revan Dai	**genehmigt von:** *authorized by*: Jeffrey Qin	X Jeffrey Qin
Datum: *Date:* 2024-10-28		**Ausstellungsdatum:** *Issue date:* 2024-10-28	
Stellung / Position:	Project Engineer	**Stellung** / Position:	Reviewer

Sonstiges / *Other*:

1. The complete test report includes the following attachments:
- Attachment 1: Equipment list (3 pages);
- Attachment 2: Photo documents (7 pages).

Zustand des Prüfgegenstandes bei Anlieferung: Prüfmuster vollständig und unbeschädigt
Condition of the test item at delivery: *Test item complete and undamaged*

* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
** Legend:*	*P(ass) = passed a.m. test specification(s)*	*F(ail) = failed a.m. test specification(s)*	*N/A = not applicable*	*N/T = not tested*

Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens.
This test report only relates to the above mentioned test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any test mark.

Anmerkungen
Remarks

1	Alle eingesetzten Prüfmittel waren zum angegebenen Prüfzeitraum gemäß eines festgelegten Kalibrierungsprogramms unseres Prüfhauses kalibriert. Sie entsprechen den in den Prüfprogrammen hinterlegten Anforderungen. Die Rückverfolgbarkeit der eingesetzten Prüfmittel ist durch die Einhaltung der Regelungen unseres Managementsystems gegeben. Detaillierte Informationen bezüglich Prüfkonditionen, Prüfequipment und Messunsicherheiten sind im Prüflabor vorhanden und können auf Wunsch bereitgestellt werden. *The equipment used during the specified testing period was calibrated according to our test laboratory calibration program. The equipment fulfils the requirements included in the relevant standards. The traceability of the test equipment used is ensured by compliance with the regulations of our management system.* *Detailed information regarding test conditions, equipment and measurement uncertainty is available in the test laboratory and could be provided on request.*
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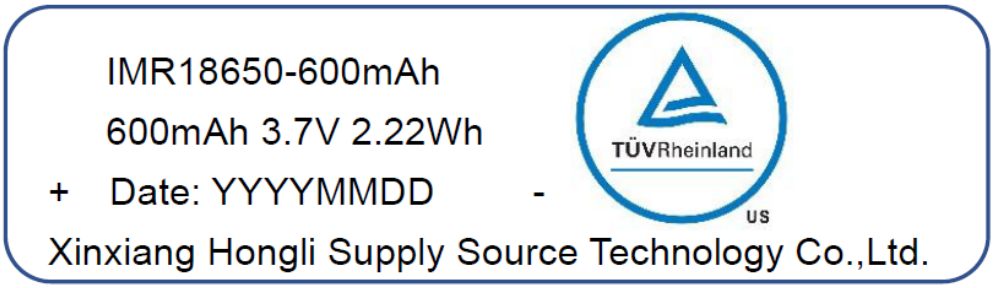
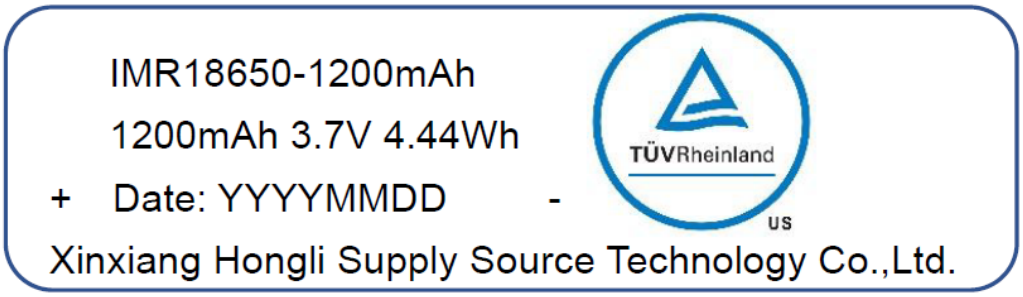
3	Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben. Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt. *Test clauses with remark of * are subcontracted to qualified subcontractors and descripted under the respective test clause in the report.* *Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.*
4	Die Entscheidungsregel für Konformitätserklärungen basierend auf numerischen Messergebnisen in diesem Prüfbericht basiert auf der "Null-Grenzwert-Regel" und der "Einfachen Akzeptanz" gemäß ILAC G8:2019 und IEC Guide 115:2021, es sei denn, in der auf Seite 1 dieses Berichts genannten angewandten Norm ist etwas anderes festgelegt oder vom Kunden gewünscht. Dies bedeutet, dass die Messunsicherheit nicht berücksichtigt wird und daher auch nicht im Prüfbericht angegeben wird. Zu weiteren Informationen bezueglich des Risikos durch diese Entscheidungsregel siehe ILAC G8:2019. *The decision rule for statements of conformity, based on numerical measurement results, in this test report is based on the "Zero Guard Band Rule" and "Simple Acceptance" in accordance with ILAC G8:2019 and IEC Guide 115:2021, unless otherwise specified in the applied standard mentioned on Page 1 of this report or requested by the customer. This means that measurement uncertainty is not taken in account and hence also not declared in the test report. For additional information to the resulting risk based of this decision rule please refer to ILAC G8:2019.*

Test item particulars:	
Information about the product needed to establish a correct test program, such as product mobility, type of power connections and similar.	(Test item particulars are selected by the TRF Originator base on the requirements in the standard)
Designation..... :	See table 4 on page 6
Trademark ..... :	N/A
Nominal voltage..... :	3.7V
Rated capacity..... :	See table 4 on page 6
Upper limit charging voltage..... :	4.2V
Maximum charge current..... :	0.5C
Manufacturer's specified end point voltage..... :	3.0V
Operating temperature range..... :	0~45°C (charge), -10~60°C (discharge)
Recommend charging method declared by the manufacturer ..... :	Charge the cell at 0.2C CC to 4.2V, then 4.2V CV until charging current reaches 0.02C at 20±5°C.
Utilization Type..... :	Technician replaceable cell
Testing:	
Test clauses and results:	
cl.10 Short-Circuit Test..... :	P
cl.11 Abnormal Charging Test..... :	P
cl.12 Forced-Discharge Test ..... :	N/A
cl.13 Crush Test ..... :	P
cl.14 Impact Test ..... :	P
cl.14A Round Bar Crush Test..... :	N/A
cl.15 Shock Test..... :	P
cl.16 Vibration Test..... :	P
cl.17 Heating Test ..... :	P
cl.18 Temperature Cycling Test ..... :	P
cl.19 Low Pressure (Altitude Simulation) Test ..... :	P
cl.20 Projectile Test..... :	P
Testing location: **CMC Testing International (Shenzhen) Co., Ltd.** 101, Building B, Kaihuimao Industrial Park, Liyuan Road, Heping Community, Fuhai Street, Baoan District, Shenzhen, Guangdong, China.	
General remarks: This report shall not be reproduced, except in full, without the written approval of the testing laboratory. The test results presented in this report relate only to the object tested. "(see remark #)" refers to a remark appended to the report. "(see appended table)" refers to a table appended to the report. Throughout this report a point is used as the decimal separator.	

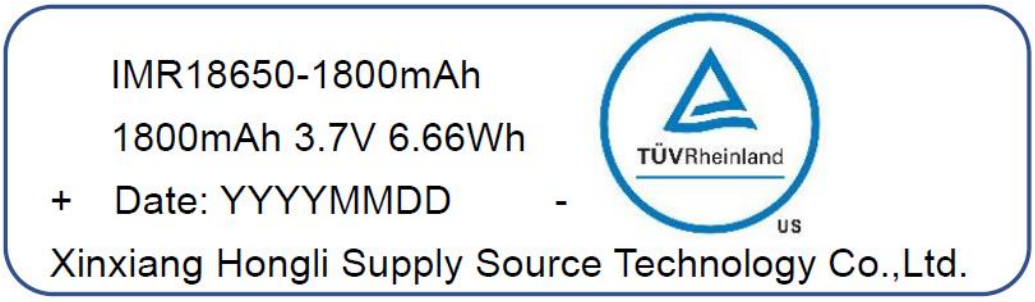
Copy of marking plate:



Label 1 for model IMR18650-600mAh



Label 2 for model IMR18650-1200mAh



Label 3 for model IMR18650-1800mAh

Remark:

1. For the date code “YYYYMMDD”:

“YYYY” represents the year of manufacture

“MM” represents the month of manufacture

“DD” represents the day of manufacture

2. Due to similarity of the marking plate, only labels of three representative models listed.

General product information:

This cell consists of the positive electrode plate, negative electrode plate, separator, electrolyte and case. The positive and negative electrode plates are housed in the case in the state being separated by the separator. The cell is evaluated according to UL 1642:2020 in this test report, and passed all single-cell tests.

1) These tested cells have not been evaluated in combination with charger(s) or host product(s). Additional evaluation to determine compliance may be required on the combination(s) in the end product evaluation.
2) The tested cells were evaluated for a maximum charge current and maximum voltage limit outlined in the Table below. The end product evaluation shall ensure that current and voltage limits noted are maintained.
3) The seven models (Model name: **IMR18650-600mAh**, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, **IMR18650-1200mAh**, IMR18650-1500mAh, **IMR18650-1800mAh**) are identical (same shape, same chemical system, using same material), except the model name and the capacity. The tested models **IMR18650-600mAh, IMR18650-1200mAh** and **IMR18650-1800mAh** are representatives of all models.
4) The charging method of cl.17: charge the cell (CC/CV) at 0.5C and 4.25V until charging current reaches 0.05C (10 pcs at 45°C and 10 pcs at -5°C).

- Table 1: Electrical parameter of model IMR18650-600mAh:

Model	Rated capacity	Nominal voltage	Nominal charge current	Nominal discharge current	Max. charge current	Max. discharge current	Upper limit charging voltage	End point voltage
IMR18650 -600mAh	600mAh	3.7V	120mA	120mA	300mA	300mA	4.2V	3.0V

- Table 1: Electrical parameter of model IMR18650-00mAh:

Model	Rated capacity	Nominal voltage	Nominal charge current	Nominal discharge current	Max. charge current	Max. discharge current	Upper limit charging voltage	End point voltage
IMR18650 -1200mAh	1200mAh	3.7V	240mA	240mA	600mA	600mA	4.2V	3.0V

- Table 1: Electrical parameter of model IMR18650-1800mAh:

Model	Rated capacity	Nominal voltage	Nominal charge current	Nominal discharge current	Max. charge current	Max. discharge current	Upper limit charging voltage	End point voltage
IMR18650 -1800mAh	1800mAh	3.7V	360mA	360mA	900mA	900mA	4.2V	3.0V

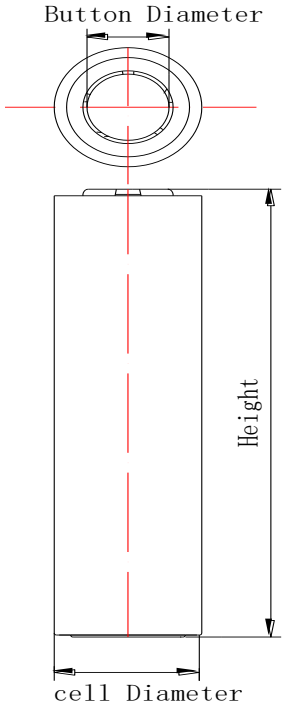
Remark: for other models feature, see Table 4 below.

- Table 4: All of the models:

Model	Energy density (Wh/L)	Rated capacity (mAh)	Max. Diameter (mm)	Max. Height (mm)	Nominal charge current (mA)	Nominal discharge current (mA)	Max. charge current (mA)	Max. discharge current (mA)
IMR18650-600mAh	**126.09**	**600**	**18.5**	**65.5**	**120**	**120**	**300**	**300**
IMR18650-700mAh	147.10	700	18.5	65.5	140	1400	350	350
IMR18650-800mAh	168.12	800	18.5	65.5	160	160	400	400
IMR18650-1050mAh	220.66	1050	18.5	65.5	210	210	525	525
IMR18650-1200mAh	**252.18**	**1200**	**18.5**	**65.5**	**240**	**240**	**600**	**600**
IMR18650-1500mAh	315.22	1500	18.5	65.5	300	300	750	750
IMR18650-1800mAh	**378.27**	**1800**	**18.5**	**65.5**	**360**	**360**	**900**	**900**

Remark: All the models have the same nominal voltage 3.7V and the same end discharge voltage 3.0V, and have the same Max. charge voltage 4.2V.

Construction:



(Dimensions value refer to Table 4)
Cell

Factory:
Xinxiang Hongli Supply Source Technology Co.,Ltd.
West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City, Henan, P.R. China

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
INTRODUCTION			
1	**Scope**		P
2	**General**		P
3	**Glossary**		P
CONSTRUCTION			
4	**General**		N/A
4.1	Casing		N/A
4.1.1	The casing of a lithium battery shall have the strength and rigidity necessary to resist the abuses to which it may be subjected, without resulting in a risk of fire. The casing of a user-replaceable lithium battery shall have the strength and rigidity necessary to resist the abuses to which it may be subjected without resulting in a risk of injury to persons.	Technician replaceable cell without enclosure.	N/A
4.1.2	A cell of a user-replaceable battery shall be in a rigid casing of sufficient strength to prevent flexing. A tool providing the mechanical advantage of a pliers, screwdriver, or hacksaw shall be the minimum capable of opening the user-replaceable cell casing, if opening of the casing will expose metallic lithium.	See above.	N/A
4.2	Electrolyte		N/A
4.2.1	A user-replaceable battery shall not contain pressurized vapor or liquid that could spray materials into the eyes or leak more than 5 mL of liquid when the battery casing is punctured under normal laboratory conditions, 23 ±2°C (73 ±3.6°F).	Technician-replaceable cell.	N/A
4.3	Use		N/A
4.3.1	A lithium battery shall be protected from abnormal charging currents during use. A battery tested and found acceptable for the charging current, IC (see Section 11), under fault conditions specified by the manufacturer, shall be protected from larger charging currents in the end product application by: a) Two blocking components, such as diodes, or b) One blocking component and one current limiting component, such as a resistor or a fuse. The current limiting component shall limit the charging current to one-third the value used in the Abnormal Charging Test, Section 11.	Single cell only.	N/A
PERFORMANCE			
5	**General**		P
5.1	Technician-replaceable Batteries		P
5.1.1	Technician-replaceable lithium cells or batteries are to be tested as described in Sections 10 – 20. Section 12, Forced-Discharge Test, is applicable only to cells intended to be used in series-connected multicell applications such as battery packs. For multicell installations, also see 5.3.1.		P

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
5.1.2	When a fire or explosion occurs as a result of the Crush Test, Section 13, or the Impact Test, Section 14, or the Round Bar Crush Test, Section 14A, or the cell or battery ruptures to the extent that the aluminum test cage is penetrated during the Projectile Test, Section 20; the use of the technician replaceable cell or battery shall be restricted to applications in which it is not exposed to, or is protected from, any conditions shown to cause a fire or explosion.	No fire or explosion occurs.	P
5.1.3	Cells and batteries subjected to the Shock Test, Section 15, Vibration Test, Section 16, Temperature Cycling Test, Section 18, and Low Pressure (Altitude Simulation) Test, Section 19, shall also not leak or vent. For these tests, unacceptable leakage is determined to have occurred when the resulting mass loss exceeds the values shown in Table 5.1, Venting and leakage mass loss criteria.	No leak or vent.	P
5.2	User-replaceable Batteries		N/A
5.2.1	User-replaceable lithium cells or batteries are to be tested as described in Sections 10 – 20. Section 12, Forced Discharge Test, is applicable only to cells intended to be used in multicell applications such as battery packs. In addition to complying with the requirements for a technician replaceable cell or battery as specified in 5.1.1, a user-replaceable cell or battery shall not explode or ignite when subjected to the Crush Test, Section 13, or the Impact Test, Section14. A user-replaceable battery shall comply with the requirements for Sections 10 – 20 and with the applicable construction requirements outlined in Section 4. Secondary lithium cells shall not be considered user-replaceable.	Technician-replaceable cell.	N/A
5.2.2	Sets of five specimens each are to be used for the Projectile Test, Section 20.3; see Table 6.1. When only one specimen from a set of five does not comply with the requirements, another set of five specimens is to be tested. All specimens from this second set shall comply with the requirements.	Technician-replaceable cell.	N/A
5.3	Multicell Installations		N/A
5.3.1	A technician-replaceable or user-replaceable cell intended for use in multicell installations or battery packs shall also be tested as described in 10.3 and Section 12. No fire or explosion shall occur as a result of these tests. In addition, batteries subjected to the test described in 10.3 shall meet the requirements as described in 5.1.1 and 5.2.1 for a cell or battery subjected to the Short-Circuit Test, Section10.	Single cell.	N/A
6	**Samples**		P

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
6.1	Fully charged primary cells or batteries and primary cells or batteries that have been conditioned by partial or complete discharge, or both, are to be used for the tests described in Sections 10 – 20. The number of samples to be used in each test for a primary cell or battery is shown in Table 6.1. When a group of cells or batteries of different sizes, but similar chemistries is involved, selected sizes representative of the range are to be tested.	The samples are secondary cells.	N/A
6.2	Fully charged secondary cells or batteries and secondary cells or batteries that have been conditioned by charge-discharge cycling are to be used for the tests described in Sections 10 – 20. The number of samples to be used in each test for a secondary cell or battery is shown in Table 6.2. When a group of cells or batteries of different sizes, and similar chemistries is involved, selected sizes representative of the range are to be tested.	The samples are secondary technician-replaceable cells. Prepared as required.	P
6.3	Prior to conducting the testing in Section 17, the lithium ion cell samples shall be pre-conditioned as outlined in 6.4 and 6.5.	Prepared as required.	P
6.4	For the heating test of Section 17, two sets of five lithium ion cell samples are to be fully discharged (i.e. to the manufacturer's specified end point voltage). The samples are then placed in a test chamber and conditioned for 1 to 4 h (5 samples at the upper temperature limit and 5 samples at the lower temperature limit of the operating region) as outlined in Table 6.3.	Prepared as required.	P
6.5	While still in the test chamber set at the temperature limits, the samples are charged (5 samples at the upper temperature limit and 5 samples at lower temperature limit) at the specified maximum charging current and upper limit charging voltage per Table 6.3, using a constant voltage charging method. Charging is continued until the charge current is reduced to the specified end of charge conditions (i.e. 0.05 times the charge current).	Prepared as required.	P
7	**Conditioning of Samples**		P
7.1	Discharge		N/A
7.1.1	Primary batteries are to be completely discharged by connecting their terminals through resistors that provide the desired level of discharge within 60 days. Batteries are to be discharged at room temperature. Cells with a liquid cathode such as thionyl chloride or sulfur dioxide, shall also be conditioned by one-half discharge	The samples are secondary cells.	N/A

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
7.1.2	For solid electrolyte and other types of primary lithium batteries that cannot be discharged within 60 days because of the small currents they inherently produce, longer discharge times plus discharge at higher temperatures may be used to obtain the desired level of discharge. The manufacturer's recommended discharge procedures are to be followed so as to obtain the required discharge level in the minimum time.	The samples are secondary cells.	N/A
7.2	Charge-discharge cycling		P
7.2.1	Secondary cells are to be conditioned at 25°C (77°F). Cells are continuously cycled as per the manufacturer's specifications. The specification shall be such that the full rated capacity of the cell is utilized and the number of cycles accumulated shall be at least equal to 25% of the advertised cycle life of the cell or cycled continuously for 90 days, whichever is shorter. Cycling is to be done either individually or in groups. Cells are to be recharged prior to testing as indicated in Table 6.2.	The samples are cycled at manufacturer's factory before they were sent for test.	P
8	**Important test considerations**		P
8.1	Some lithium batteries are capable of exploding when the tests described in Sections 10 – 20 are conducted. It is important that personnel be protected from the flying fragments, explosive force, sudden release of heat, and noise that results from such explosions. The test area is to be well ventilated to protect personnel from possible harmful fumes or gases.	Prepared the tests as required.	P
8.2	As an additional precaution, the temperatures on the surface of the battery casings shall be monitored during the tests described in Sections 10, 11, 12, 13, and 14. All personnel involved in the testing of lithium batteries are to be instructed never to approach a lithium battery while the surface temperature exceeds 90°C (194°F) and not to touch the lithium battery while the surface temperature exceeds 45°C (113°F).	Prepared the tests as required.	P
8.3	For protection, the Projectile Test, Section 20 is to be conducted in a room separate from the observer.	Prepared the tests as required.	P
9	**Temperature Measurements**		P
9.1	Temperatures are to be measured by thermocouples consisting of wires not larger than 24 AWG (0.21 mm2) and not smaller than 30 AWG (0.05 mm2) and a potentiometer-type instrument.	Prepared the tests as required.	P
9.2	The temperature measurements on the batteries are to be made with the measuring junction of the thermocouple held tightly against the metal casing of the battery.	Prepared the tests as required. Casing temperature was recorded on the center of the cell surface.	P
TESTS FOR TECHNICIAN-REPLACEABLE AND USER-REPLACEABLE BATTERIES			
ELECTRICAL TESTS			
10	**Short-Circuit Test**		P

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
10.1	Each test sample battery, in turn, is to be short-circuited by connecting the positive and negative terminals of the battery with a circuit load having a resistance load of 80±20 mΩ. The temperature of the battery case is to be recorded during the test. The battery is to discharge until a fire or explosion is obtained, or until it has reached a completely discharged state of less than 0.2 V and the battery case temperature has returned to ±10°C (±18°F) of ambient temperature. The voltage at the end of the test may not reach 0.2 V due to operation of protective devices in the circuit. The return to near ambient of the battery (cell) casing in an indication of ultimate results.	Tested as required. See table 10.	P
10.2	Tests are to be conducted at 20 ±5°C (68 ±9°F) and at 55 ±5°C (131 ±9°F). The batteries are to reach equilibrium at 20 ±5°C or 55 ±5°C, as applicable, before the terminals are connected.	Tested as required.	P
10.3	A battery is to be tested individually unless the manufacturer indicates that it is intended for use in series or parallel. For series or parallel use, additional tests on five sets of batteries are to be conducted using the maximum number of batteries to be covered for each configuration.	Tested as required.	P
10.4	When an overcurrent protective device activates during the test, the test shall be repeated with the battery supply connected to the maximum load that does not cause the protective device to open. Protective devices that are relied upon to meet the compliance criteria for the short circuit test shall comply with 2.3.1.	Only one single lithium-ion cell, no over-current or thermal protective device was integrated into the cell.	N/A
10.5	The samples shall not explode or catch fire.	The test results meet the requirements.	P
11	**Abnormal Charging Test**		P
11.1	Primary cells or batteries shall comply with 11.2 – 11.7.	Secondary cell.	N/A
11.2	Cells or batteries conditioned in accordance with Tables 6.1, as applicable, are to be used for this test. The batteries are to be tested in an ambient temperature of 20 ±5°C (68 ±9°F).		N/A

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
11.3	Each test sample battery is to be subjected to a charging current of three times the current Ic, specified by the manufacturer by connecting it in opposition to a dc-power supply. The specified charging current is to be obtained by connecting a resistor of the specified size and rating in series with the battery. The test charging time is to be calculated using the formula: $t_c = \frac{2.5C}{3(I_c)}$, in which tc is the charging time in hour C is the capacity of the cell/battery in ampere-hours, and Ic is the maximum charging current, in amperes, specified by the manufacturer. The minimum charging time is to be 7 hours.		N/A
11.4	When a non-resettable overcurrent or protective device operates during the test, the test is to be repeated at a charge current below the level that the protective device operates. When a resettable protective device operates during the test, the protector is allowed to reset to a total of 10 cycles; or until the appropriate charging time has been completed, but not less than 7 hours. Protective devices that are relied upon to meet the compliance criteria for the abnormal charging test shall comply with 2.3.1.		N/A
11.5	The samples shall not explode or catch fire.		N/A
11.6	Secondary cells or batteries shall comply with 11.7 – 11.10.	See table 11, tested as required.	P
11.7	Cells or batteries conditioned in accordance with Tables 6.2, as applicable, are to be used for this test. The batteries are to be tested in an ambient temperature of 20 ±5°C (68 ±9°F).	Tested as required.	P
11.8	Each test sample battery is to be discharged at a constant current of 0.2 C/1 h, to a manufacturer specified discharge endpoint voltage. The cell or battery is then to be charged with a constant maximum specified output voltage and a current limit of three times the maximum charging current Ic, specified by the manufacturer. Charging duration is to be 7 hours or the time required to reach the manufacturer's specified end-of-charge condition, whichever is greater.	Tested as required.	P

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
11.9	When a non-resettable overcurrent or protective device operates during the test, the test shall be repeated at an overcharging current below the level that the protection device operates. When a resettable protective device operates during the test, the protector is to be allowed to reset to a total of 10 cycles; or until the appropriate charging time has been completed, but not less than 7 hours. Protective devices that are relied upon to meet the compliance criteria for the abnormal charging test shall comply with 2.3.1.	The samples are to be tested without any assistance of over-current or thermal protective devices.	N/A
11.10	The samples shall not explode or catch fire.	No explosion or catch fire during and after the test.	P
12	**Forced-Discharged Test**		N/A
12.1	This test is intended for cells that are to be used in series-connected, multicell applications, such as battery packs.	One single cell.	N/A
12.2	A fully discharged cell is to be force-discharged by connecting it in series with fully charged cells of the same kind. The number of fully charged cells to be connected in series with the discharged cell is to equal the maximum number less one of the cells to be covered for series use. Five cells are to be completely discharged, at room temperature.		N/A
12.3	Once the fully discharged cell is connected in series with the specified number of fully charged cells the resultant battery pack is to be short circuited.		N/A
12.4	The positive and negative terminals of the sample are to be connected with a copper wire with a resistance load of 80±20mΩ. The sample is to discharge until a fire or explosion is obtained, or until it has reached a completely discharged state of less than 0.2V and the battery case temperature has returned to ±10°C (18°F) of ambient temperature. The voltage at the end of the test may not reach 0.2V due to operation of protective devices in the circuit. The return to near ambient of the cell casing is an indication of ultimate results.		N/A
12.5	When an overcurrent or protective operates during the test, the test shall be repeated with the battery supply connected to the maximum load that does not cause the protective device to open. Protective devices that are relied upon to meet the compliance criteria for the forced discharge test shall comply with 2.3.1.		N/A
12.6	The samples shall not explode or catch fire.		N/A
MECHANICAL TESTS			
13	**Crush Test**		P

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
13.1	A battery is to be crushed between two flat surfaces. The force for the crushing is to be applied by a hydraulic ram or similar force mechanism. The flat surfaces are to be brought in contact with the cells and the crushing is to be continued until an applied force of 13 ±1 kN (3000 ±224 lbs) is reached. Once the maximum force has been obtained it is to be released.	Tested as required. See table 13.	P
13.2	A cylindrical, pouch or prismatic cell is to be crushed with its longitudinal axis parallel to the flat surfaces of the crushing apparatus. A prismatic cell is also to be rotated 90° around its longitudinal axis so that both the wide and narrow sides will be subjected to the crushing force. Each sample is to be subjected to a crushing force in only one direction. Separate samples are to be used for each test.	Tested as required. The samples are cylindrical lithium-ion cells.	P
	Exception: For Lithium ion systems, a cylindrical, pouch or prismatic cell is to be crushed with its longitudinal axis parallel to the flat surface of the crushing apparatus. Each sample is to be subjected to a crushing force in only one direction. Test only the wide side of pouch and prismatic cells.	Li-ion cell.	P
13.3	A coin or button battery is to be crushed with the flat surface of the battery parallel with the flat surfaces of the crushing apparatus.	Not a coin or button battery.	N/A
13.4	The samples shall not explode or catch fire.	No explosion or catch fire during and after the test.	P
14	**Impact Test**		P
14.1	A test sample battery is to be placed on a flat surface. A 15.8 ±0.1-mm (5/8 ±0.004-in) diameter bar is to be placed across the center of the sample. A 9.1 ±0.46-kg (20 ±1-lb) weight is to be dropped from a height of 610 ±25 mm (24 ±1 in) onto the sample. See Figure 14.1.	Tested as required. See table 14.	P
14.2	A cylindrical, pouch or prismatic cell is to be impacted with its longitudinal axis parallel to the flat surface and perpendicular to the longitudinal axis of the 15.8-mm (5/8-in) diameter curved surface lying across the center of the test sample. A prismatic cell is also to be rotated 90° around its longitudinal axis so that both the wide and narrow sides are subjected to the impact. Each sample is to be subjected to only a single impact. Separate samples are to be used for each test.	Tested as required.	P
	Exception 1: For Lithium ion systems, a cylindrical, pouch or prismatic cell is to be impacted with its longitudinal axis parallel to the flat surface and perpendicular to the longitudinal axis of the 15.8-mm (5/8-in) diameter curved surface lying across the center of the test sample. Each sample is to be subjected to only a single impact. Test only the wide side of pouch and prismatic cells.	The samples are lithium-ion cells.	P

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Clause	Requirement + Test	Result - Remark	Verdict
	Exception 2: Pouch cells not evaluated to this test, and with a capacity greater than 300 mAh, shall comply with the Round Bar Crush Test in Section 14A.		P
14.3	A coin or button battery is to be impacted with the flat surface of the test sample parallel to the flat surface and the 15.8-mm (5/8-in) diameter curved surface lying across its center.	Not a coin or button battery.	N/A
14.4	The samples shall not explode or catch fire.	No explosion or catch fire.	P
14A	**Round Bar Crush Test**		N/A
14A.1	With reference to Exception No. 2 in 14.2, pouch cells not evaluated with the Impact Test in Section 14 and with a capacity greater than 300 mAh shall be evaluated with this test.		N/A
14A.2	The sample shall be fully charged with the method recommended by the manufacturer.		N/A
14A.3	The sample shall be positioned on a flat surface with a 25 ±1 mm (1 ±0.039 in) diameter steel round bar placed on top of the sample. The edge of the bar is to be aligned with the top edge of the cell body, with the longitudinal axis of the bar perpendicular to the tab of the cell. The bar shall extend beyond the width of the test sample on each side by at least 5 mm (0.197 in) as shown in Figure 14A.1. A preload pressure of 0.4 – 0.6 N/cm^2 (0.58 – 0.87 psi) shall be applied before the sample thickness is measured in 14A.5 and the crush force in 14A.6 is applied. See 14A.4. NOTE: A sample may be taped, glued or clamped to the flat surface, avoiding obstructing the round bar placement and crush plate motion, to eliminate any unevenness between the sample and the flat test surface.		N/A
14A.4	For cells with the positive tab and negative tab on opposite sides, the bar placement in 14A.3 and the crush force application in 14A.6 are applied on each side of the cell near the tabs, using a different set of samples for each side.		N/A
14A.5	After the bar placement per 14A.3 or 14A.4 and the preload pressure is applied per 14A.3, the thickness of the sample shall be measured allowing a tolerance of ±0.1 mm (±0.039 in), using the Measuring Method of Cell Thickness in the Annex for Dimensions of the Cell with a Laminate Film Case in the Standard for Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes – Secondary Lithium Cells and Batteries for Portable Applications – Part 3: Prismatic and Cylindrical Lithium Secondary Cells, and Batteries Made from Them, IEC 61960-3.		N/A

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
14A.6	A crush force shall be applied onto the round bar, with a crushing direction of 90 ±1° between the crushing direction and round bar and the crushing direction and test platform. The displacement in the vertical direction shall be measured. The moving speed of the crush plate is to be no greater than 0.1 mm/s (0.004 in/s). The displacement of the crush plate shall be stopped and held for 30 s once the cell deformation reaches 13 ±1% of the cell thickness (starting from the preload thickness measured in 14A.5) or the crush force reaches the force value in Table 14A.1, whichever comes first. See Figure 14A.1 and Figure 14A.2.		N/A
14A.7	The samples shall not explode or catch fire.		N/A
15	**Shock Test**		P
15.1	The cell is to be secured to the testing machine by means of a rigid mount which supports all mounting surfaces of the cell. Each cell shall be subjected to a total of three shocks of equal magnitude. The shocks are to be applied in each of three mutually perpendicular directions unless it has only two axes of symmetry in which case only two directions shall be tested. Each shock is to be applied in a direction normal to the face of the cell. For each shock the cell is to be accelerated in such a manner that during the initial 3 ms the minimum average acceleration is 75 g (where g is the local acceleration due to gravity). The peak acceleration shall be between 125 and 175 g. Cells shall be tested at a temperature of 20 ± 5°C (68 ± 9°F).	Prepared the test as required. See table 15.	P
15.2	The samples shall not explode or catch fire. In addition, the sample shall not vent or leak as described in 5.1.1.	No explosion or catch fire, the sample not vent or leak.	P
16	**Vibration Test**		P
16.1	A battery is to be subjected to simple harmonic motion with an amplitude of 0.8 mm (0.03 inch) [1.6 mm (0.06 inch) total maximum excursion].	Prepared the test as required. See table 16.	P
16.2	The frequency is to be varied at the rate of 1 Hz/min between 10 and 55 Hz, and return in not less than 90 nor more than 100 min. The battery is to be tested in three mutually perpendicular directions. For a battery that has only two axes of symmetry, the battery is to be tested perpendicular to each axis.	Tested as required.	P
16.3	The samples shall not explode or catch fire. In addition the sample shall not vent or leak as described in 5.1.1.	The test results meet the requirements.	P
ENVIRONMENTAL TESTS			
17	**Heating Test**		P

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
17.1	A battery is to be heated in a gravity convection or circulating air oven with an initial temperature of 20 ±5°C (68±9°F). The temperature of the oven is to be raised at a rate of 5 ±2°C (9±3.6°F) per minute to a temperature of 130 ±2°C (266±3.6°F) and remain for 10 min. The sample shall return to room temperature (20±5°C) and then be examined. For batteries specified for temperatures above 100°C (212°F), the conditioning temperature shall be increased from 130 ±2°C (266±3.6°F), to 30±2°C (86±3.6°F) above the manufacturers maximum specified temperature. For a battery of lithium metal chemistry, the conditioning temperature shall be increased to a maximum of 170 ±2°C (338±3.6°F).	Tested as required. Oven temperature: 130°C.	P
17.2	The samples shall not explode or catch fire.	The test results meet the requirements.	P
18	**Temperature Cycling Test**		P
18.1	The batteries are to be placed in a test chamber and subjected to the following cycles: a) Raising the chamber-temperature to 70 ±3°C (158 ±5°F) within 30 minutes and maintaining this temperature for 4 hours. b) Reducing the chamber temperature to 20 ±3°C (68 ±5°F) within 30 minutes and maintaining this temperature for 2 hours. c) Reducing the chamber temperature to minus 40 ±3°C (minus 40 ±5°F) within 30 minutes and maintaining this temperature for 4 hours. d) Raising the chamber temperature to 20 ±3°C (68 ±5°F) within 30 minutes. e) Repeating the sequence for a further 9 cycles. f) After the 10th cycle, storing the batteries for a minimum of 24 hours, at a temperature of 20 ±5°C (68 ±9°F) prior to examination.	Tested as required. See table 18.	P
18.2	The samples shall not explode or catch fire. In addition, the samples shall not vent or leak as described in 5.1.1.	The test results meet the requirements.	P
19	**Low Pressure (Altitude Simulation) Test**		P
19.1	Sample batteries are to be stored for 6 hours at an absolute pressure of 11.6 kPa (1.68 psi) and a temperature of 20 ± 3°C (68 ± 5°F).	Tested as required. See table 19.	P
19.2	The samples shall not explode or catch fire as a result of the Low Pressure (Altitude Simulation) Test. In addition, the samples shall not vent or leak as described in 5.1.1.	The test results meet the requirements.	P
FIRE EXPOSURE TEST			
20	**Projectile Test**		P
20.1	When subjected to the test described in 20.2 - 20.5 no part of an exploding cell or battery shall penetrate the wire screen such that some or all of the cell or battery protrudes through the screen.	The test results meet the requirements.	P

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
20.2	Each test sample cell or battery is to be placed on a screen that covers a 102-mm (4-inch) diameter hole in the center of a platform table. The screen is to be constructed of steel wire mesh having 20 openings per 25.4 mm (1 in) and a wire diameter of 0.43 mm (0.017 in).	Prepared the test as required.	P
20.3	The screen is to be mounted 38 mm (1-1/2 in) above a Meker type burner. The fuel and air flow rates are to be set to provide a bright blue flame that causes the supporting screen to glow a bright red.	Prepared the test as required.	P
20.4	An eight-sided covered wire cage, 610-mm (2-ft) across and 305-mm (1-ft) high, made from metal screening is to be placed over the test sample. See Figure 20.1. The metal screening is to be constructed from 0.25-mm (0.010-in) diameter aluminum wire with 16 -18 wires per 25.4 mm (1 inch) in each direction.	Tested as required.	P
20.5	The sample is to be heated and shall remain on the screen until it explodes or the cell or battery has ignited and burned out. It is not required to secure the sample in place unless it is at risk of falling off the screen before the test is completed. When required, the sample shall be secured to the screen with a single wire tied around the sample.	Tested as required.	P
MARKING			
21	**General**		P
21.1	A battery shall be legibly and permanently marked with: The manufacturer's name, trade name, or trademark or other descriptive marking by which the organization responsible for the product may be identified; A distinctive ("catalog" or "model") number or the equivalent; The date or other dating period of manufacture not exceeding any three consecutive months.	See marking plate on page 4.	P
21.2	If a manufacturer produces a battery at more than one factory, each battery package shall have a distinctive marking to identify it as the product of a particular factory.	One factory only.	N/A
22	**Primary Batteries**		N/A
22.1	A primary battery shall be marked with the word "WARNING" and the following or an equivalent statement: "Risk of fire and burns. Do not recharge, open, crush, heat above (the manufacturer's specified temperature rating), or incinerate." If space does not permit marking on the battery, the marking may be on the smallest unit package.	Secondary cell.	N/A

UL 1642:2020			
Clause	Requirement + Test	Result - Remark	Verdict
22.2	The packaging for a user-replaceable battery shall be marked with the word "CAUTION" and the following or equivalent statements: "Risk of fire and burns. Do not recharge, disassemble, heat above (the manufacturer's specified temperature rating), or incinerate. Keep battery out of reach of children and in original package until ready to use. Dispose of used batteries promptly."		N/A
22.3	For user replaceable lithium primary coin cells (3.0 V) the packaging shall also include the following or equivalent: "WARNING – Never put batteries in mouth. Swallowing may lead to serious injury or death. If ingested, immediately seek medical attention and have the doctor phone the National Capital Poison Control Center." This marking may be combined with the marking of 22.2, if the signal word "WARNING" is used instead of "CAUTION."		N/A

10	TABLE: Short-Circuit Test (model: IMR18650-1800mAh)				P
Fully Charged Samples					
Sample No.	SN240813012 C001	SN240813012 C002	SN240813012 C003	SN240813012 C004	SN24081301 2C005
Ambient temperature: (at 20±5°C, °C)	23.1	23.1	23.1	23.1	23.1
Max. casing temp Tmax (°C)	126.7	125.1	124.9	128.4	128.2
Sample No.	SN240813012 C006	SN240813012 C007	SN240813012 C008	SN240813012 C009	SN24081301 2C010
Ambient temperature: (at 55 ±5°C, °C)	57.7	57.7	57.7	57.7	57.7
Max. casing temp Tmax (°C)	122.6	125.1	124.2	124.9	123.3
Cycled Samples					
Sample No.	SN240813012 C011	SN240813012 C012	SN240813012 C013	SN240813012 C014	SN24081301 2C015
Ambient temperature: (at 20±5°C, °C)	23.1	23.1	23.1	23.1	23.1
Max. casing temp Tmax (°C)	121.8	120.3	120.1	120.6	120.4
Sample No.	SN240813012 C016	SN240813012 C017	SN240813012 C018	SN240813012 C019	SN24081301 2C020
Ambient temperature: (at 55 ±5°C, °C)	57.7	57.7	57.7	57.7	57.7
Max. casing temp Tmax (°C)	117.7	121.2	121.0	119.8	118.3
Supplementary information: Tmax was recorded on the centre of the cell surface. No explode or catch fire.					

10	TABLE: Short-Circuit Test (model: IMR18650-1200mAh)				P
Fully Charged Samples					
Sample No.	SN240813013 C001	SN240813013 C002	SN240813013 C003	SN240813013 C004	SN24081301 3C005
Ambient temperature: (at 20±5°C, °C)	23.0	23.0	23.0	23.0	23.0
Max. casing temp Tmax (°C)	128.3	130.0	123.0	120.8	129.2
Sample No.	SN240813013 C006	SN240813013 C007	SN240813013 C008	SN240813013 C009	SN24081301 3C010
Ambient temperature: (at 55 ±5°C, °C)	58.0	58.0	58.0	58.0	58.0
Max. casing temp Tmax (°C)	121.1	122.8	121.1	126.4	123.7
Cycled Samples					
Sample No.	SN240813013 C011	SN240813013 C012	SN240813013 C013	SN240813013 C014	SN24081301 3C015
Ambient temperature: (at 20±5°C, °C)	23.0	23.0	23.0	23.0	23.0
Max. casing temp Tmax (°C)	130.4	124.9	127.5	122.8	121.6
Sample No.	SN240813013 C016	SN240813013 C017	SN240813013 C018	SN240813013 C019	SN24081301 3C020
Ambient temperature: (at 55 ±5°C, °C)	58.0	58.0	58.0	58.0	58.0
Max. casing temp Tmax (°C)	113.3	115.4	112.2	111.2	119.1
Supplementary information: Tmax was recorded on the centre of the cell surface. No explode or catch fire.					

10	TABLE: Short-Circuit Test (model: IMR18650-600mAh)				P
Fully Charged Samples					
Sample No.	SN240813014 C001	SN240813014 C002	SN240813014 C003	SN240813014 C004	SN24081301 4C005
Ambient temperature: (at 20±5°C, °C)	21.5	21.5	21.5	21.5	21.5
Max. casing temp Tmax (°C)	127.7	128.4	127.9	127.6	123.1
Sample No.	SN240813014 C006	SN240813014 C007	SN240813014 C008	SN240813014 C009	SN24081301 4C010
Ambient temperature: (at 55 ±5°C, °C)	58.1	58.1	58.1	58.1	58.1
Max. casing temp Tmax (°C)	114.3	115.7	115.6	116.1	117.3
Cycled Samples					
Sample No.	SN240813014 C011	SN240813014 C012	SN240813014 C013	SN240813014 C014	SN24081301 4C015
Ambient temperature: (at 20±5°C, °C)	21.5	21.5	21.5	21.5	21.5
Max. casing temp Tmax (°C)	121.3	123.5	124.5	121.0	120.6
Sample No.	SN240813014 C016	SN240813014 C017	SN240813014 C018	SN240813014 C019	SN24081301 4C020
Ambient temperature: (at 55 ±5°C, °C)	58.1	58.1	58.1	58.1	58.1
Max. casing temp Tmax (°C)	111.5	109.8	112.9	112.7	112.3
Supplementary information: Tmax was recorded on the centre of the cell surface. No explode or catch fire.					

11	TABLE: Abnormal Charging Test (model: IMR18650-1800mAh)				P
Id	0.36 A		Ue	3.0 V	
Ic	0.9 A		Uc	4.2 V	
Fully Charged Samples					
Sample No.	SN240813012 C021	SN240813012 C022	SN240813012 C023	SN240813012 C024	SN24081301 2C025
Ambient temperature: (°C)	21.2	21.2	21.2	21.2	21.2
Max. casing temp Tmax (°C)	29.8	30.3	30.5	30.2	30.3
Cycled Samples					
Sample No.	SN240813012 C026	SN240813012 C027	SN240813012 C028	SN240813012 C029	SN24081301 2C030
Ambient temperature: (°C)	21.2	21.2	21.2	21.2	21.2
Max. casing temp Tmax (°C)	31.9	32.0	31.8	31.6	30.1
Supplementary information: Test current is 2.7A. No explode or catch fire.					

11	TABLE: Abnormal Charging Test (model: IMR18650-1200mAh)				P
Id	0.24 A		Ue	3.0 V	
Ic	0.6 A		Uc	4.2 V	
Fully Charged Samples					
Sample No.	SN240813013 C021	SN240813013 C022	SN240813013 C023	SN240813013 C024	SN24081301 3C025
Ambient temperature: (°C)	22.4	22.4	22.4	22.4	22.4
Max. casing temp Tmax (°C)	31.5	31.2	31.4	31.6	31.2
Cycled Samples					
Sample No.	SN240813013 C026	SN240813013 C027	SN240813013 C028	SN240813013 C029	SN24081301 3C030
Ambient temperature: (°C)	22.4	22.4	22.4	22.4	22.4
Max. casing temp Tmax (°C)	31.5	30.9	32.5	32.9	32.3
Supplementary information: Test current is 1.8A. No explode or catch fire.					

11	TABLE: Abnormal Charging Test (model: IMR18650-600mAh)				P
Id	0.12 A		Ue	3.0 V	
Ic	0.3 A		Uc	4.2 V	
Fully Charged Samples					
Sample No.	SN240813014 C021	SN240813014 C022	SN240813014 C023	SN240813014 C024	SN24081301 4C025
Ambient temperature: (°C)	22.4	22.4	22.4	22.4	22.4
Max. casing temp Tmax (°C)	29.0	27.3	27.0	27.2	27.3
Cycled Samples					
Sample No.	SN240813014 C026	SN240813014 C027	SN240813014 C028	SN240813014 C029	SN24081301 4C030
Ambient temperature: (°C)	22.4	22.4	22.4	22.4	22.4
Max. casing temp Tmax (°C)	28.0	27.9	28.0	27.6	28.0
Supplementary information: Test current is 0.9A. No explode or catch fire.					

13	TABLE: Crush Test (model: IMR18650-1800mAh)				P
Ambient temperature: 23.1°C					
Fully Charged Samples					
Sample No.	SN240813012 C031	SN240813012 C032	SN240813012 C033	SN240813012 C034	SN24081301 2C035
Fire or explosion?	No	No	No	No	No
Crush direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Cycled Samples					
Sample No.	SN240813012 C036	SN240813012 C037	SN240813012 C038	SN240813012 C039	SN24081301 2C040
Fire or explosion?	No	No	No	No	No
Crush direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Supplementary information: no explosion or catch fire.					

13	TABLE: Crush Test (model: IMR18650-1200mAh)				P
Ambient temperature: 23.6°C					
Fully Charged Samples					
Sample No.	SN240813013 C031	SN240813013 C032	SN240813013 C033	SN240813013 C034	SN24081301 3C035
Fire or explosion?	No	No	No	No	No
Crush direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Cycled Samples					
Sample No.	SN240813013 C036	SN240813013 C037	SN240813013 C038	SN240813013 C039	SN24081301 3C040
Fire or explosion?	No	No	No	No	No
Crush direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Supplementary information: no explosion or catch fire.					

13	TABLE: Crush Test (model: IMR18650-600mAh)				P
Ambient temperature: 22.4°C					
Fully Charged Samples					
Sample No.	SN240813014 C031	SN240813014 C032	SN240813014 C033	SN240813014 C034	SN24081301 4C035
Fire or explosion?	No	No	No	No	No
Crush direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Cycled Samples					
Sample No.	SN240813014 C036	SN240813014 C037	SN240813014 C038	SN240813014 C039	SN24081301 4C040
Fire or explosion?	No	No	No	No	No
Crush direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Supplementary information: no explosion or catch fire.					

14	TABLE: Impact Test (model: IMR18650-1800mAh)				P
Ambient temperature: 22.6°C					
Fully Charged Samples					
Sample No.	SN240813012 C041	SN240813012 C042	SN240813012 C043	SN240813012 C044	SN24081301 2C045
Fire or explosion?	No	No	No	No	No
Impact direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Cycled Samples					
Sample No.	SN240813012 C046	SN240813012 C047	SN240813012 C048	SN240813012 C049	SN24081301 2C050
Fire or explosion?	No	No	No	No	No
Impact direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Supplementary information: no explosion or catch fire.					

14	TABLE: Impact Test (model: IMR18650-1200mAh)				P
Ambient temperature: 22.5°C					
Fully Charged Samples					
Sample No.	SN240813013 C041	SN240813013 C042	SN240813013 C043	SN240813013 C044	SN24081301 3C045
Fire or explosion?	No	No	No	No	No
Impact direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Cycled Samples					
Sample No.	SN240813013 C046	SN240813013 C047	SN240813013 C048	SN240813013 C049	SN24081301 3C050
Fire or explosion?	No	No	No	No	No
Impact direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Supplementary information: no explosion or catch fire.					

14	TABLE: Impact Test (model: IMR18650-600mAh)				P
Ambient temperature: 23.3°C					
Fully Charged Samples					
Sample No.	SN240813014 C041	SN240813014 C042	SN240813014 C043	SN240813014 C044	SN24081301 4C045
Fire or explosion?	No	No	No	No	No
Impact direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Cycled Samples					
Sample No.	SN240813014 C046	SN240813014 C047	SN240813014 C048	SN240813014 C049	SN24081301 4C050
Fire or explosion?	No	No	No	No	No
Impact direction	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface	Longitudinal axis parallel to the flat surface
Supplementary information: no explosion or catch fire.					

14A	TABLE: Round Bar Crush Test				N/A
Ambient temperature: °C					
Fully Charged Samples					
Sample No.					
Fire or explosion?					
Cell thickness (mm)					
Cycled Samples					
Sample No.					
Fire or explosion?					
Cell thickness (mm)					
Supplementary information:					

15	TABLE: Shock Test (model: IMR18650-1800mAh)				P
Ambient temperature: 22.0°C					
Fully Charged Samples					
Sample No.	SN240813012 C051	SN240813012 C052	SN240813012 C053	SN240813012 C054	SN24081301 2C055
Mass before test (g)	42.024	42.021	42.018	42.016	42.020
Mass after test (g)	42.023	42.020	42.016	42.014	42.020
Mass loss ratio (%)	0.002	0.002	0.005	0.005	0.000
Cycled Samples					
Sample No.	SN240813012 C056	SN240813012 C057	SN240813012 C058	SN240813012 C059	SN24081301 2C060
Mass before test (g)	42.019	42.026	42.014	42.016	42.023
Mass after test (g)	42.017	42.025	42.013	42.015	42.022
Mass loss ratio (%)	0.005	0.002	0.002	0.002	0.002
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

15	TABLE: Shock Test (model: IMR18650-1200mAh)				P
Ambient temperature: 22.5°C					
Fully Charged Samples					
Sample No.	SN240813013 C051	SN240813013 C052	SN240813013 C053	SN240813013 C054	SN24081301 3C055
Mass before test (g)	34.076	34.070	34.077	34.071	34.064
Mass after test (g)	34.075	34.070	34.076	34.068	34.062
Mass loss ratio (%)	0.003	0.000	0.003	0.009	0.006
Cycled Samples					
Sample No.	SN240813013 C056	SN240813013 C057	SN240813013 C058	SN240813013 C059	SN24081301 3C060
Mass before test (g)	34.070	34.066	34.068	34.063	34.065
Mass after test (g)	34.070	34.064	34.067	34.062	34.063
Mass loss ratio (%)	0.000	0.006	0.003	0.003	0.006
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

15	TABLE: Shock Test (model: IMR18650-600mAh)				P
Ambient temperature: 22.5°C					
Fully Charged Samples					
Sample No.	SN240813014 C051	SN240813014 C052	SN240813014 C053	SN240813014 C054	SN24081301 4C055
Mass before test (g)	26.034	26.031	26.046	26.041	26.043
Mass after test (g)	26.032	26.030	26.044	26.040	26.041
Mass loss ratio (%)	0.008	0.004	0.008	0.004	0.008
Cycled Samples					
Sample No.	SN240813014 C056	SN240813014 C057	SN240813014 C058	SN240813014 C059	SN24081301 4C060
Mass before test (g)	26.035	26.030	26.041	26.040	26.043
Mass after test (g)	26.035	26.030	26.040	26.040	26.042
Mass loss ratio (%)	0.000	0.000	0.004	0.000	0.004
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

16	TABLE: Vibration Test (model: IMR18650-1800mAh)				P
Ambient temperature: 22.6°C					
Fully Charged Samples					
Sample No.	SN240813012 C061	SN240813012 C062	SN240813012 C063	SN240813012 C064	SN24081301 2C065
Mass before test (g)	42.019	42.026	42.014	42.020	42.026
Mass after test (g)	42.018	42.024	42.011	42.019	42.025
Mass loss ratio (%)	0.002	0.005	0.007	0.002	0.002
Cycled Samples					
Sample No.	SN240813012 C066	SN240813012 C067	SN240813012 C068	SN240813012 C069	SN24081301 2C070
Mass before test (g)	42.018	42.015	42.011	42.024	42.021
Mass after test (g)	42.017	42.014	42.010	42.023	42.020
Mass loss ratio (%)	0.002	0.002	0.002	0.002	0.002
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

16	TABLE: Vibration Test (model: IMR18650-1200mAh)				P
Ambient temperature: 22.6°C					
Fully Charged Samples					
Sample No.	SN240813013 C061	SN240813013 C062	SN240813013 C063	SN240813013 C064	SN24081301 3C065
Mass before test (g)	34.071	34.069	34.062	34.067	34.074
Mass after test (g)	34.070	34.067	34.060	34.067	34.073
Mass loss ratio (%)	0.003	0.006	0.006	0.000	0.003
Cycled Samples					
Sample No.	SN240813013 C066	SN240813013 C067	SN240813013 C068	SN240813013 C069	SN24081301 3C070
Mass before test (g)	34.071	34.068	34.066	34.069	34.061
Mass after test (g)	34.070	34.067	34.065	34.069	34.060
Mass loss ratio (%)	0.003	0.003	0.003	0.000	0.003
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

16	TABLE: Vibration Test (model: IMR18650-600mAh)				P
Ambient temperature: 22.6°C					
Fully Charged Samples					
Sample No.	SN240813014 C061	SN240813014 C062	SN240813014 C063	SN240813014 C064	SN24081301 4C065
Mass before test (g)	26.036	26.041	26.033	26.039	26.030
Mass after test (g)	26.032	26.040	26.033	26.037	26.029
Mass loss ratio (%)	0.015	0.004	0.000	0.008	0.004
Cycled Samples					
Sample No.	SN240813014 C066	SN240813014 C067	SN240813014 C068	SN240813014 C069	SN24081301 4C070
Mass before test (g)	26.032	26.034	26.035	26.038	26.036
Mass after test (g)	26.032	26.033	26.035	26.038	26.035
Mass loss ratio (%)	0.000	0.004	0.000	0.000	0.004
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

18	TABLE: Temperature Cycling Test (model: IMR18650-1800mAh)				P
Ambient temperature: 23.5°C					
Fully Charged Samples					
Sample No.	SN240813012 C091	SN240813012 C092	SN240813012 C093	SN240813012 C094	SN24081301 2C095
Mass before test (g)	42.026	42.019	42.015	42.017	42.023
Mass after test (g)	41.988	41.982	41.980	41.989	42.001
Mass loss ratio (%)	0.090	0.088	0.083	0.067	0.052
Cycled Samples					
Sample No.	SN240813012 C096	SN240813012 C097	SN240813012 C098	SN240813012 C099	SN24081301 2C100
Mass before test (g)	42.023	42.020	42.024	42.021	42.021
Mass after test (g)	41.991	41.994	41.987	42.000	41.991
Mass loss ratio (%)	0.076	0.062	0.088	0.050	0.071
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

18	TABLE: Temperature Cycling Test (model: IMR18650-1200mAh)				P
Ambient temperature: 23.3°C					
Fully Charged Samples					
Sample No.	SN240813013 C091	SN240813013 C092	SN240813013 C093	SN240813013 C094	SN24081301 3C095
Mass before test (g)	34.076	34.073	34.070	34.070	34.075
Mass after test (g)	34.047	34.043	34.051	34.044	34.045
Mass loss ratio (%)	0.085	0.088	0.056	0.076	0.088
Cycled Samples					
Sample No.	SN240813013 C096	SN240813013 C097	SN240813013 C098	SN240813013 C099	SN24081301 3C100
Mass before test (g)	34.069	34.070	34.067	34.066	34.068
Mass after test (g)	34.042	34.045	34.041	34.036	34.039
Mass loss ratio (%)	0.079	0.073	0.076	0.088	0.085
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

18	TABLE: Temperature Cycling Test (model: IMR18650-600mAh)				P
Ambient temperature: 23.2°C					
Fully Charged Samples					
Sample No.	SN240813014 C091	SN240813014 C092	SN240813014 C093	SN240813014 C094	SN24081301 4C095
Mass before test (g)	26.030	26.044	26.039	26.037	26.034
Mass after test (g)	26.016	26.021	26.026	26.020	26.014
Mass loss ratio (%)	0.054	0.088	0.050	0.065	0.077
Cycled Samples					
Sample No.	SN240813014 C096	SN240813014 C097	SN240813014 C098	SN240813014 C099	SN24081301 4C100
Mass before test (g)	26.036	26.035	26.031	26.035	26.032
Mass after test (g)	26.022	26.019	26.011	26.020	26.012
Mass loss ratio (%)	0.054	0.061	0.077	0.058	0.077
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

19	TABLE: Low Pressure (Altitude Simulation) Test (model: IMR18650-1800mAh)				P
Ambient temperature: 22.1°C					
Fully Charged Samples					
Sample No.	SN240813012 C101	SN240813012 C102	SN240813012 C103	SN240813012 C104	SN24081301 2C105
Mass before test (g)	42.014	42.018	42.022	42.027	42.019
Mass after test (g)	42.013	42.017	42.020	42.025	42.017
Mass loss ratio (%)	0.002	0.002	0.005	0.005	0.005
Cycled Samples					
Sample No.	SN240813012 C106	SN240813012 C107	SN240813012 C108	SN240813012 C109	SN24081301 2C110
Mass before test (g)	42.014	42.021	42.020	42.016	42.025
Mass after test (g)	42.013	42.020	42.020	42.015	42.025
Mass loss ratio (%)	0.002	0.002	0.000	0.002	0.000
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

19	TABLE: Low Pressure (Altitude Simulation) Test (model: IMR18650-1200mAh)				P
Ambient temperature: 22.1°C					
Fully Charged Samples					
Sample No.	SN240813013C101	SN240813013C102	SN240813013C103	SN240813013C104	SN240813013C105
Mass before test (g)	34.069	34.077	34.073	34.075	34.070
Mass after test (g)	34.067	34.075	34.072	34.074	34.070
Mass loss ratio (%)	0.006	0.006	0.003	0.003	0.000
Cycled Samples					
Sample No.	SN240813013C106	SN240813013C107	SN240813013C108	SN240813013C109	SN240813013C110
Mass before test (g)	34.074	34.068	34.071	34.075	34.072
Mass after test (g)	34.073	34.067	34.070	34.075	34.071
Mass loss ratio (%)	0.003	0.003	0.003	0.000	0.003
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

19	TABLE: Low Pressure (Altitude Simulation) Test (model: IMR18650-600mAh)				P
Ambient temperature: 22.1°C					
Fully Charged Samples					
Sample No.	SN240813014C101	SN240813014C102	SN240813014C103	SN240813014C104	SN240813014C105
Mass before test (g)	26.036	26.045	26.039	26.033	26.043
Mass after test (g)	26.034	26.044	26.037	26.032	26.041
Mass loss ratio (%)	0.008	0.004	0.008	0.004	0.008
Cycled Samples					
Sample No.	SN240813014C106	SN240813014C107	SN240813014C108	SN240813014C109	SN240813014C110
Mass before test (g)	26.040	26.037	26.043	26.041	26.044
Mass after test (g)	26.039	26.035	26.041	26.040	26.042
Mass loss ratio (%)	0.004	0.008	0.008	0.004	0.008
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%.					

Critical Components					
Material: e.g. external enclosure, PCB, closed-end connector, sleeves, cord anchorage etc					
Components with winding: e.g. motor, transformer, magnetic coil etc.					
Other components: e.g. switch, thermostat, heater, plug, internal wire, capacitor, relay, varistor etc.					
Object/ Part No.	**Manufacturer/ trademark**	**Type/ Model**	**Technical data**	**Standard**	**Mark(s) of conformity**
Cell	Xinxiang Hongli Supply Source Technology Co.,Ltd.	1) IMR18650-600mAh 2) IMR18650-700mAh 3) IMR18650-800mAh 4) IMR18650-1050mAh 5) IMR18650-1200mAh 6) IMR18650-1500mAh 7) IMR18650-1800mAh	1) 3.7V, 600mAh 2) 3.7V, 700mAh 3) 3.7V, 800mAh 4) 3.7V, 1050mAh 5) 3.7V, 1200mAh 6) 3.7V, 1500mAh 7) 3.7V, 1800mAh	UL 1642:2020 R10.22	Tested with appliance
-Positive electrode	Xinxiang Hongli Energy Co., Ltd.	HL-04	$LiMn_2O_4$	--	--
-Negative electrode	Yunnan Zhongsheng New Materials Co., Ltd	AG-1	Graphite	--	--
-Separator	Henan Huiqiang New Energy Materials Technology Co., Ltd.	PP 0.025×61mm	PP, Shutdown temperature: 130°C	--	--
-Electrolyte	Xinxiang Huarui Lithium New Energy Co., Ltd.	HR-HL01	$LiPF_6$, EC, DEC, EMC, Conductivity: 10.5mS/cm, H2O<20ppm	--	--
-Cell case	Xinxiang Shengda New Energy Technology Co., Ltd.	18650	Low Carbon Steel, 0.22mm± 0.02mm	--	--

--End of main report--

Test location:

CMC Testing International (Shenzhen) Co., Ltd.

101, Building B, Kaihuimao Industrial Park, Liyuan Road, Heping Community, Fuhai Street, Baoan District, Shenzhen, Guangdong, China.

Main Test Apparatus

	Item	Equipment	Model	Technical Parameter	Cal.	Due.
☑	CMC-YQ-001	Programmable Temp.& Humi Chamber	ZZ-K01B	(-70°C ~ 150°C) 20%RH ~ 98%RH	2024-03-20	2025-03-19
☑	CMC-YQ-002	Programmable Temp.& Humi Chamber	ZZ-K01B	(-70°C ~ 150°C) 20%RH ~ 98%RH	2024-03-20	2025-03-19
☑	CMC-YQ-003	Battery Test System	ACTS-20V5A	20V 5A	2024-03-20	2025-03-19
□	CMC-YQ-004	Battery Test System	PBTS-20V5A	20V 5A	2024-03-20	2025-03-19
☑	CMC-YQ-005	Battery Test System	ACTS-20V10A	20V 10A	2024-03-20	2025-03-19
☑	CMC-YQ-006	Battery Extrusion Testing machine	GX-5067-TSM	1-10V 1-20KN (13KN) (-20°C ~ 100°C)	2024-03-20	2025-03-19
☑	CMC-YQ-007	Battery Burning Testing Machine	GX-6053	/	2024-03-20	2025-03-19
☑	CMC-YQ-008	Battery thermal shock tester	SPH101	(5±2)°C/min Ambient temperature +20°C ~ +200°C	2024-03-20	2025-03-19
□	CMC-YQ-009	Battery forced internal short circuit testing machine	GX-6055-CSM	1-10V, 1-1000N (50N/250N/400N/800N) (-5°C ~ 100°C)	2024-03-20	2025-03-19
☑	CMC-YQ-010	Battery short-circuit tester	GX-6055-B5HL	100V, 100A 80 mΩ ± 20 mΩ	2024-03-20	2025-03-19
☑	CMC-YQ-011	Battery weight impact testing machine	ZZ-A11	9.1kg/10kg 15.8mm (5/8inch) 610mm/1000mm	2024-03-20	2025-03-19
☑	CMC-YQ-012	Low Altitude Simulation Tester	GX-3020-ZC80	11.6KPa 0.5 ~ 100KPa	2024-03-20	2025-03-19
□	CMC-YQ-013	Single-Blade Drop Tester	GX-6050-A	300mm ~ 1500mm	2024-03-20	2025-03-19
☑	CMC-YQ-014	Vibration test instrument	EV206VT650	Maximum load: 220Kg Frequency range: moving coil: 5-2500Hz, Exciting force: sine: 5880N (600Kg.f), Vertical table: 128m/s2 (15g), water table: 118m/s2 (12g)	2024-03-20	2025-03-19
☑	CMC-YQ-015	Vertical Shock Test Instrument	SKT50	30g-175g, 6ms-11ms	2024-03-20	2025-03-19

	Item	Equipment	Model	Technical Parameter	Cal.	Due.
☑	CMC-YQ-016-1	Data Collector	34970A	0°C ~ 300°C	2024-03-20	2025-03-19
☑	CMC-YQ-016-2	Data Collector	34970A	0°C ~ 300°C	2024-03-20	2025-03-19
□	CMC-YQ-016-3	Data Collector	34970A	0°C ~ 300°C	2024-03-20	2025-03-19
□	CMC-YQ-019	DC POWER	PSW80	80V, 43A	2024-03-20	2025-03-19
□	CMC-YQ-019-1	DC POWER	PSW80-40.5	80V, 43A	2024-03-20	2025-03-19
□	CMC-YQ-021	Electronic platform scale	TCS-200KG	(0~200kg)±1g	2024-03-20	2025-03-19
☑	CMC-YQ-022	Electronic Balance	JCS-6=5103W	(0~500g)±0.001g	2024-03-20	2025-03-19
□	CMC-YQ-023	Digital caliper	MNT-150	(0-150) mm	2024-03-20	2025-03-19
□	CMC-YQ-024	Vernier caliper	(0~200)mm	(0~200)mm	2024-03-20	2025-03-19
□	CMC-YQ-025	Digital Multimeter	FLUKE-17B+	40mV ~ 1000 V 40μA ~ 10 A 4Ω ~ 40 MΩ	2024-03-20	2025-03-19
□	CMC-YQ-026	Dc Low Internal Testing Resistance	KJ2511	10μΩ ~ 20KΩ	2024-03-20	2025-03-19
□	CMC-YQ-027	Ac Internal Testing Resistance	JK2520B	0.001mΩ ~ 33Ω 0.0001~60.000V	2024-03-20	2025-03-19
□	CMC-YQ-028	Programmed DC Electronic Load	SJ8512B	0V-150V 0.1mA-60A	2024-03-20	2025-03-19
☑	CMC-YQ-029	Timer	XL-009A	9H ~ 1/100s	2024-03-13	2025-03-12
□	CMC-YQ-031	Battery Test System	RCDS-100V100A	0V ~ 100V -100A ~ 100A Minimum discharge voltage: 7V Minimum output current: 150mA 100A	2024-03-20	2025-03-19
□	CMC-YQ-032	Insulation resistance tester	UT501A	1000V 0.0MΩ-5.5GΩ	2024-03-20	2025-03-19
□	CMC-YQ-034	Dew-point Meter	S505	(-50°C ~ +20°C) Dew point±2°C	2024-03-14	2025-03-13
□	CMC-YQ-043	Swallow Gauge	/	Short side 25.4mm, long side 57.1mm, diameter 31.7mm	2024-03-20	2025-03-19
☑	CMC-YQ-044-1	Temperature & humidity monitoring	ZL-TH10TP	(-20 ~ 50°C; 0-100%RH)	2024-03-20	2025-03-19
☑	CMC-YQ-044-2	Temperature & humidity monitoring	ZL-TH10TP	(-20 ~ 50°C; 0-100%RH)	2024-03-20	2025-03-19

	Item	Equipment	Model	Technical Parameter	Cal.	Due.
☑	CMC-YQ-044-3	Temperature & humidity monitoring	ZL-TH10TP	(-20 ~ 50°C; 0-100%RH)	2024-03-20	2025-03-19
☑	CMC-YQ-044-4	Temperature & humidity monitoring	ZL-TH10TP	(-20 ~ 50°C; 0-100%RH)	2024-03-20	2025-03-19
☑	CMC-YQ-044-5	Temperature & humidity monitoring	ZL-TH10TP	(-20 ~ 50°C; 0-100%RH)	2024-03-20	2025-03-19
☑	CMC-YQ-044-6	Temperature & humidity monitoring	ZL-TH10TP	(-20 ~ 50°C; 0-100%RH)	2024-03-20	2025-03-19

Product: Cylindrical Lithium-ion Rechargeable Cell

Type Designation: IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh

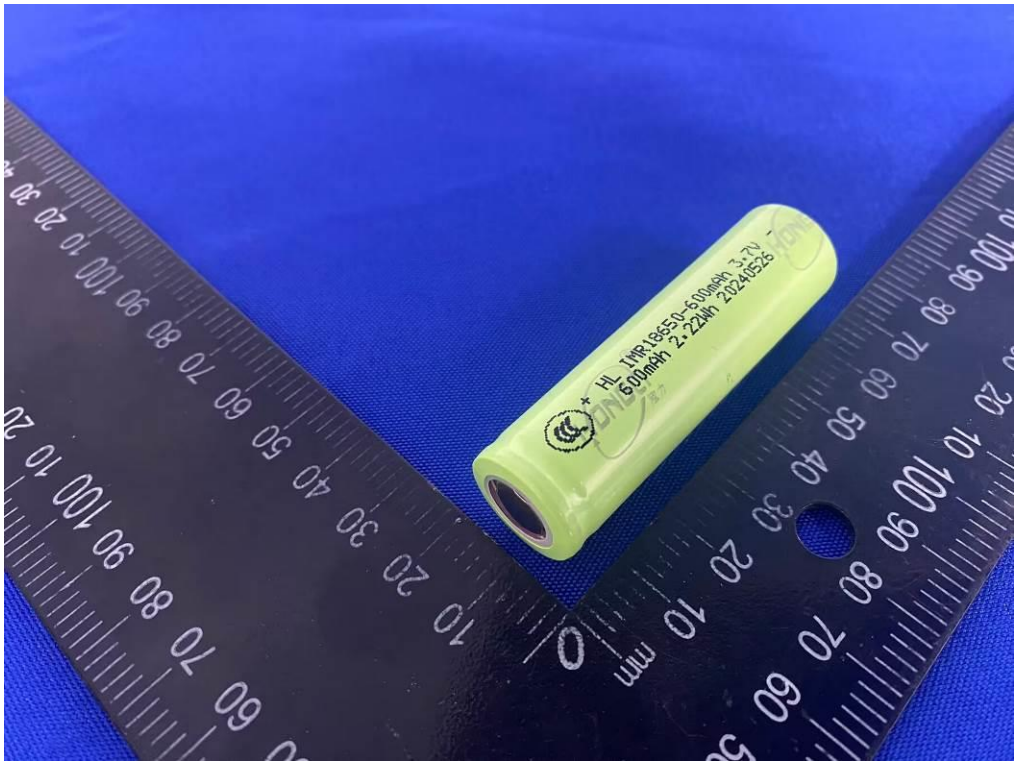

Figure 1 Side view-1 of cell for model IMR18650-600mAh

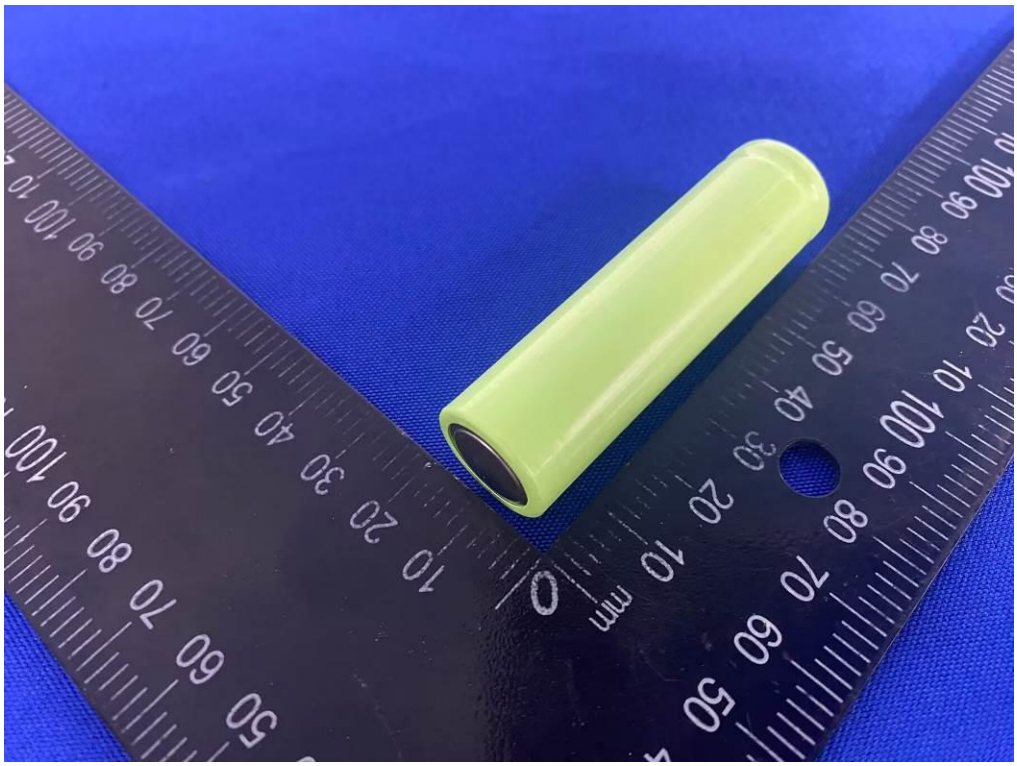

Figure 2 Side view-2 of cell for model IMR18650-600mAh

Product: Cylindrical Lithium-ion Rechargeable Cell

Type Designation: IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh

Figure 3 Side view-1 of cell for model IMR18650-700mAh

Figure 4 Side view-2 of cell for model IMR18650-700mAh

Product: Cylindrical Lithium-ion Rechargeable Cell

Type Designation: IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh

Figure 5 Side view-1 of cell for model IMR18650-800mAh

Figure 6 Side view-2 of cell for model IMR18650-800mAh

Product: Cylindrical Lithium-ion Rechargeable Cell

Type Designation: IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh

Figure 7 Side view-1 of cell for model IMR18650-1050mAh

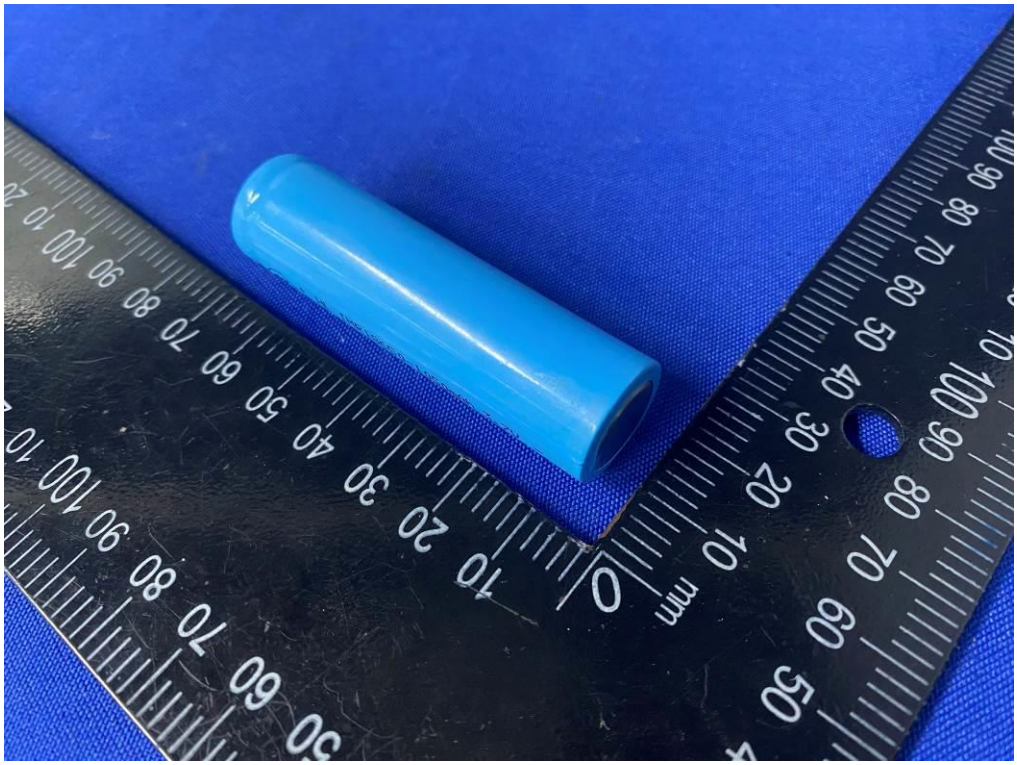

Figure 8 Side view-2 of cell for model IMR18650-1050mAh

Product: Cylindrical Lithium-ion Rechargeable Cell

Type Designation: IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh

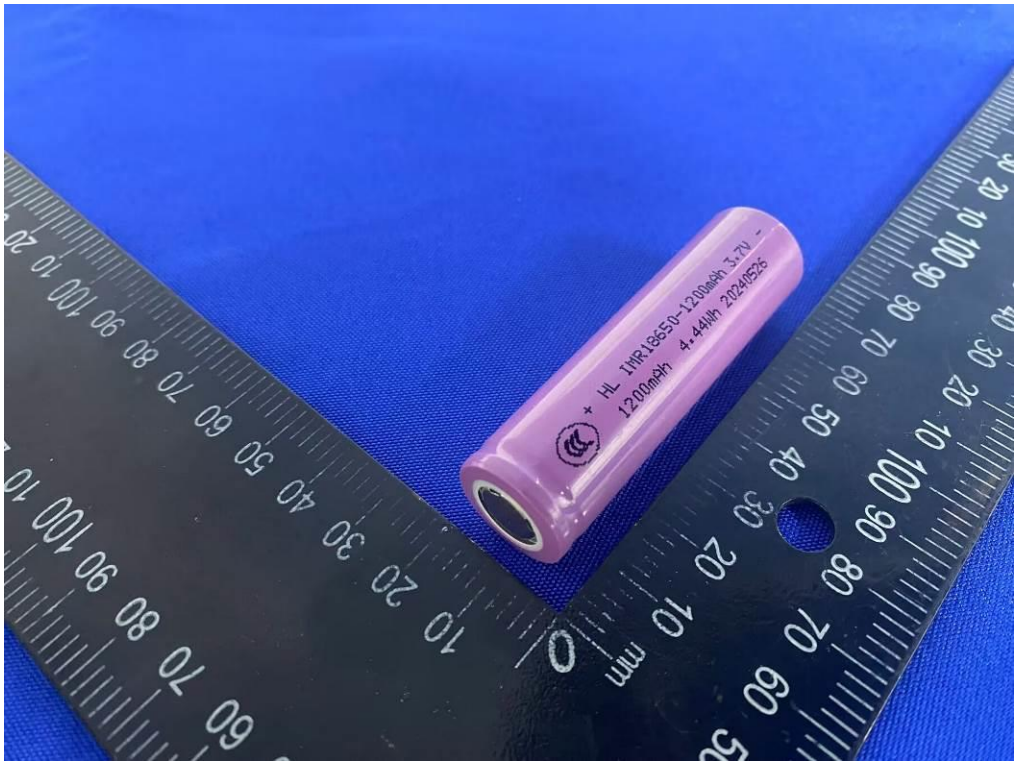

Figure 9 Side view-1 of cell for model IMR18650-1200mAh

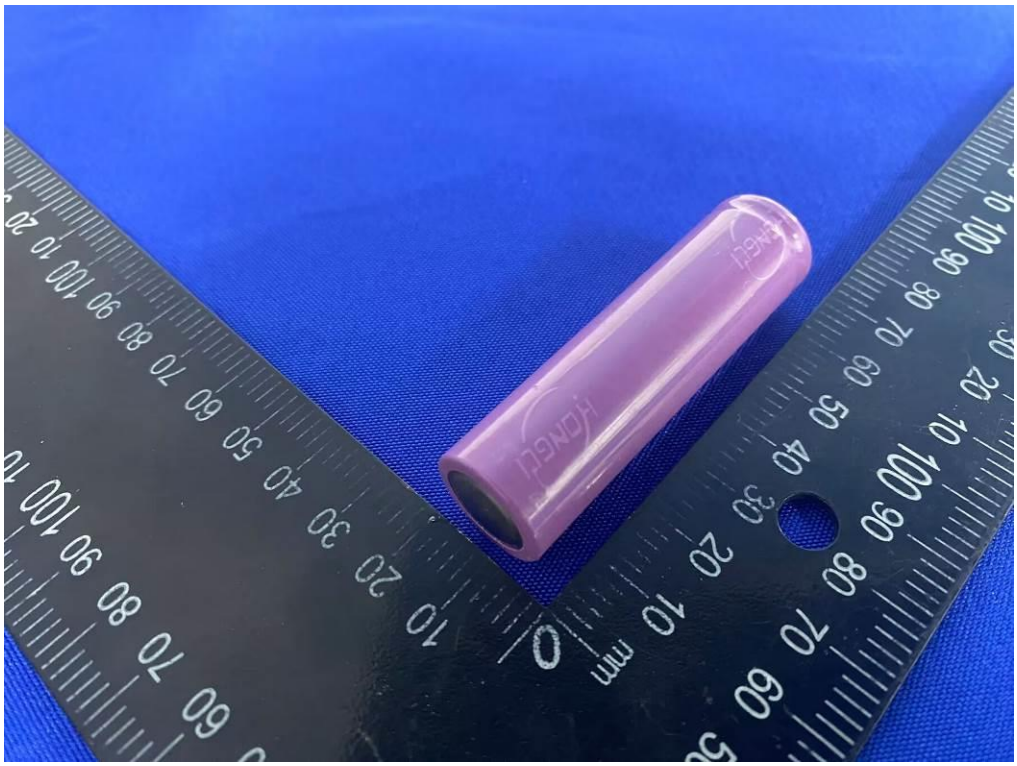

Figure 10 Side view-2 of cell for model IMR18650-1200mAh

Product: Cylindrical Lithium-ion Rechargeable Cell

Type Designation: IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh

Figure 11 Side view-1 of cell for model IMR18650-1500mAh

Figure 12 Side view-2 of cell for model IMR18650-1500mAh

Product: Cylindrical Lithium-ion Rechargeable Cell

Type Designation: IMR18650-600mAh, IMR18650-700mAh, IMR18650-800mAh, IMR18650-1050mAh, IMR18650-1200mAh, IMR18650-1500mAh, IMR18650-1800mAh

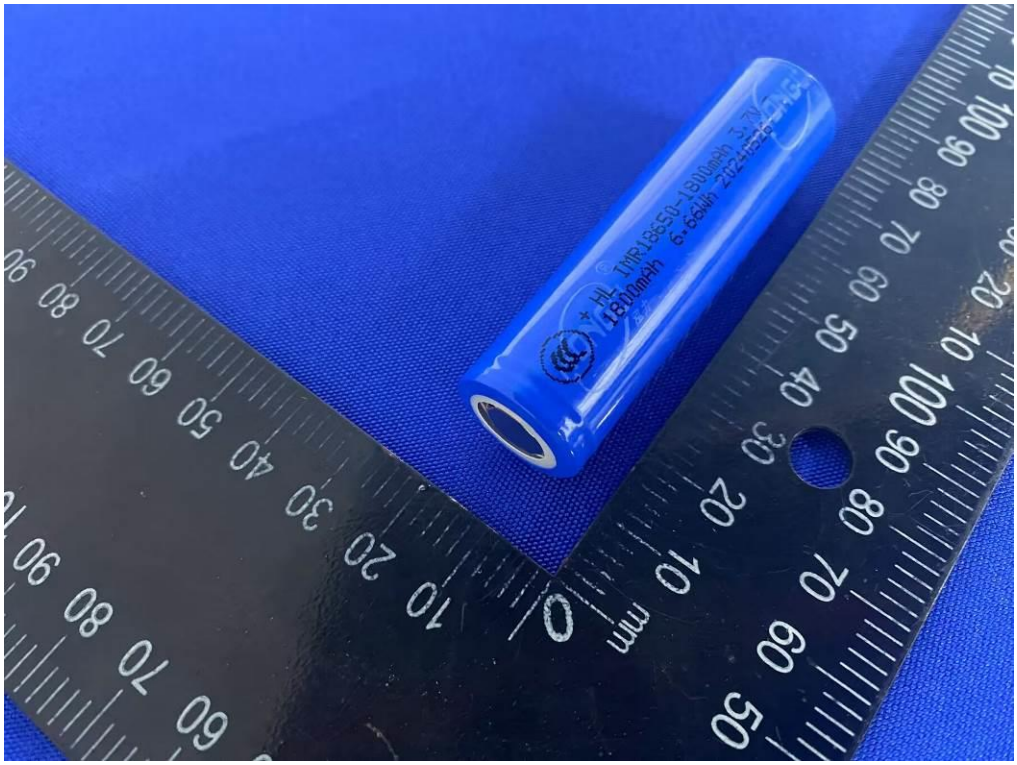

Figure 13 Side view-1 of cell for model IMR18650-1800mAh

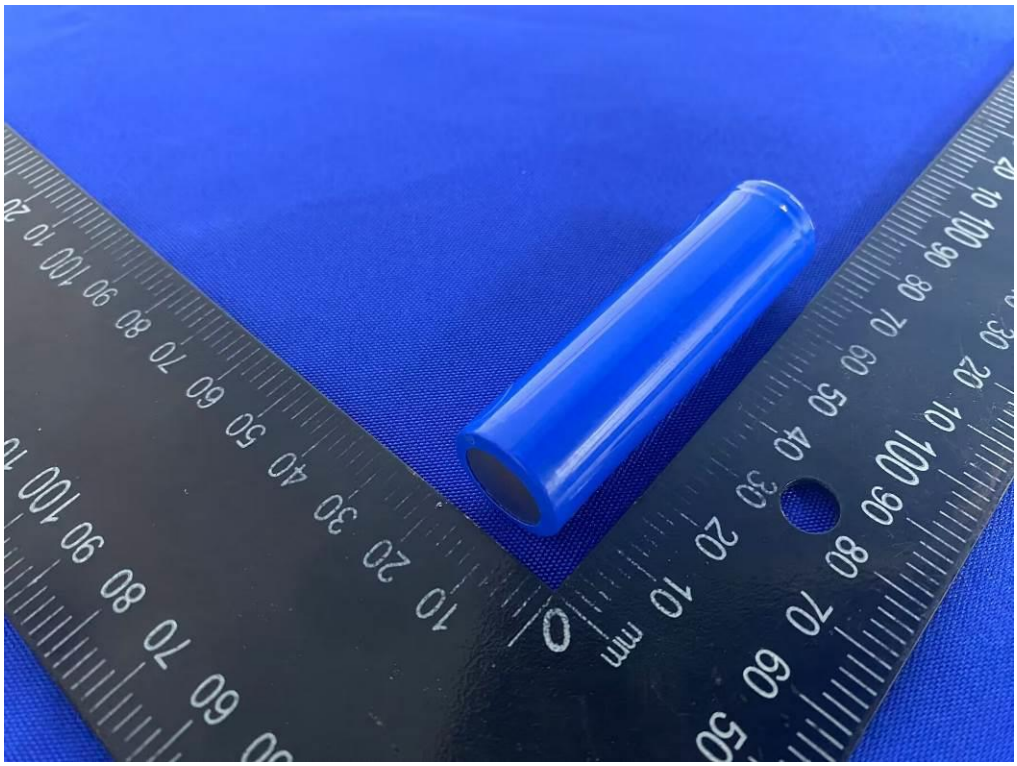

Figure 14 Side view-2 of cell for model IMR18650-1800mAh